

Geometry: Angle Measures of Intersecting Lines



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DIRECTIONS

Use angle relationships to solve for x and find missing angle measures. Show all work.

- 1 Vertical Angles — Solve for x :

$$\angle 1 = (3x + 15)^\circ, \quad \angle 3 = (5x - 9)^\circ$$

Answer: _____

- 2 Linear Pair — Solve for x :

$$\angle 1 = (4x + 5)^\circ, \quad \angle 2 = (6x - 5)^\circ$$

Answer: _____

- 3 Find the supplement of the angle:

$$73^\circ$$

Answer: _____

- 4 Vertical Angles — Find x and $m\angle A$:

$$m\angle A = (8x - 10)^\circ, \quad m\angle C = (6x + 20)^\circ$$

Answer: _____

- 5 Complementary Angles — Solve for x :

$$\angle P = (2x + 12)^\circ, \quad \angle Q = (3x - 2)^\circ$$

Answer: _____

- 6 Linear Pair — Find x and both angle measures:

$$\angle 1 = (x + 10)^\circ, \quad \angle 2 = (2x + 5)^\circ$$

Answer: _____

- 7 Two lines intersect. Find $m\angle 2$, $m\angle 3$, and $m\angle 4$:

$$m\angle 1 = 72^\circ$$

Answer: _____

- 8 Vertical Angles — Find x and $m\angle 1$:

$$m\angle 1 = (5x + 3)^\circ, \quad m\angle 2 = (7x - 9)^\circ$$

Answer: _____



Answer Key & Solutions

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TEACHER NOTES

Key rules: Vertical angles are equal. Linear pairs sum to 180° . Supplementary angles sum to 180° . Complementary angles sum to 90° .

1 Vertical Angles — Solve for x:

$$\angle 1 = (3x + 15)^\circ, \quad \angle 3 = (5x - 9)^\circ$$

$$= x = 12$$

$3x+15 = 5x-9$ (vertical angles are equal) $\rightarrow 24 = 2x \rightarrow x = 12$.

2 Linear Pair — Solve for x:

$$\angle 1 = (4x + 5)^\circ, \quad \angle 2 = (6x - 5)^\circ$$

$$= x = 18$$

Linear pair sums to 180° : $(4x+5)+(6x-5)=180 \rightarrow 10x=180 \rightarrow x=18$.

3 Find the supplement of the angle:

$$73^\circ$$

$$= 107^\circ$$

$180^\circ - 73^\circ = 107^\circ$.

4 Vertical Angles — Find x and $m\angle A$:

$$m\angle A = (8x - 10)^\circ, \quad m\angle C = (6x + 20)^\circ$$

$$= x = 15, \quad m\angle A = 110^\circ$$

$8x-10 = 6x+20 \rightarrow 2x=30 \rightarrow x=15$. $m\angle A = 8(15)-10 = 110^\circ$.

5 Complementary Angles — Solve for x:

$$\angle P = (2x + 12)^\circ, \quad \angle Q = (3x - 2)^\circ$$

$$= x = 16$$

$(2x+12)+(3x-2)=90 \rightarrow 5x+10=90 \rightarrow 5x=80 \rightarrow x=16$.

6 Linear Pair — Find x and both angle measures:

$$\angle 1 = (x + 10)^\circ, \quad \angle 2 = (2x + 5)^\circ$$

$$= x = 55, \quad \angle 1 = 65^\circ, \quad \angle 2 = 115^\circ$$

$(x+10)+(2x+5)=180 \rightarrow 3x+15=180 \rightarrow x=55$.

7 Two lines intersect. Find $m\angle 2$, $m\angle 3$, and $m\angle 4$:

$$m\angle 1 = 72^\circ$$

$$= m\angle 2 = 108^\circ, \quad m\angle 3 = 72^\circ, \quad m\angle 4 = 108^\circ$$

$\angle 3 = \angle 1$ (vertical). $\angle 2 = \angle 4 = 180^\circ - 72^\circ = 108^\circ$ (linear pairs).

8 Vertical Angles — Find x and $m\angle 1$:

$$m\angle 1 = (5x + 3)^\circ, \quad m\angle 2 = (7x - 9)^\circ$$

$$= x = 6, \quad m\angle 1 = 33^\circ$$

$5x+3=7x-9 \rightarrow 12=2x \rightarrow x=6$. $m\angle 1 = 5(6)+3 = 33^\circ$.

